

WASP-43b

R-1840

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_200501 · created 2026-08-03T06:52:14+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458970.6752 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.057
Transit depth (Rp/Rs)^2	3.88 +/- 2.25 [%]
Orbital Inclination (inc)	87.09 +/- 2.9
Airmass coefficient 1 (a1)	4.03 +/- 1.48
Airmass coefficient 2 (a2)	-0.92 +/- 0.48
Scatter in the residuals of the lightcurve fit is	34.5 %
Best Comparison Star	None
Optimal Aperture	1.84
Optimal Annulus	8.93
Transit Duration (day)	0.0611 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.057 vs 0.1594 (deviation 0.66σ)
Duration fit vs published	0.0611 d vs 0.0512 d (deviation 2.1σ)
Mid-time O–C	-3.3 min
Depth SNR	3.5
Residual scatter	34.5 %

Light curve

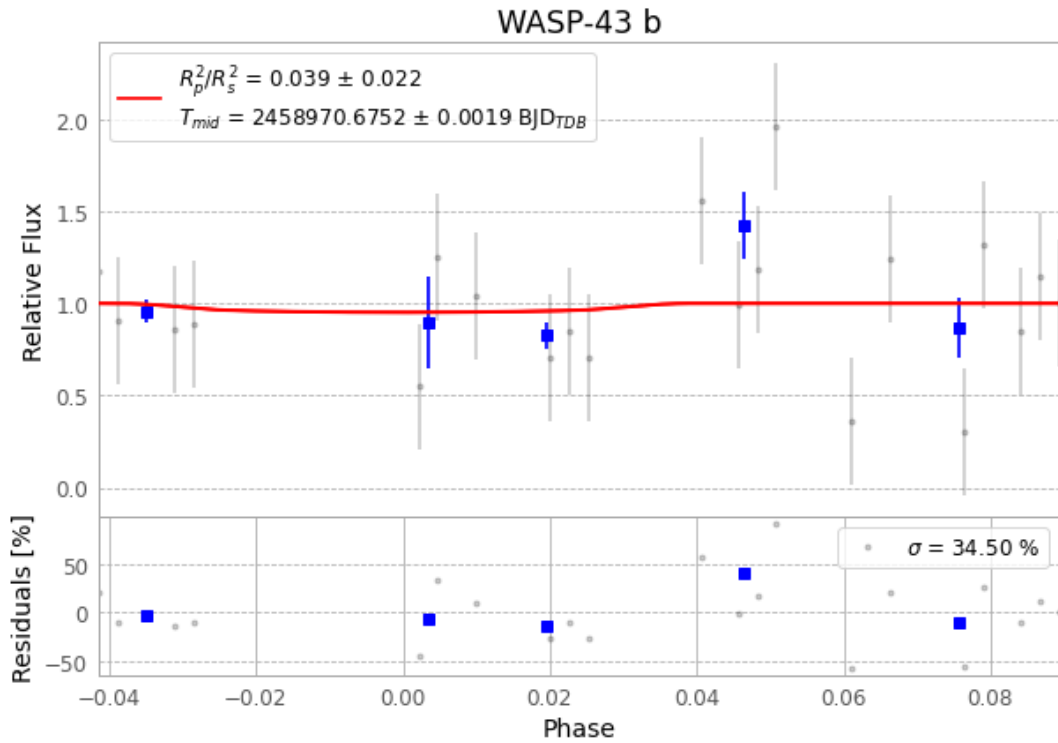


Figure 1 — FinalLightCurve_WASP-43 b_2020-04-30.png (SHA-256 27b7dd0ec94a7af6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [630, 302], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 651560413d26c5ad...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

53 FITS frames (35 MB), WASP-43200501031812.FITS → WASP-43200501055413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200501.log (04d533a8d3ef...), FinalLightCurve_WASP-43 b_2020-04-30.png (27b7dd0ec94a...), FinalLightCurve_WASP-43 b_2020-04-30.csv (4aa7b15a3eb6...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 06:52Z.